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Dual contrastive learning with graph masking: A self-supervised framework for multi-view clustering

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ABSTRACT

Graph-based deep multi-view clustering has recently emerged as a prominent research paradigm, driven by its efficacy in modeling nonlinear feature relationships. However, two critical limitations persist in existing approaches: (1) They exhibit high susceptibility to noise and redundancy, which often distorts the graph construction process; and (2) Excessive emphasis on cross-view feature alignment inadvertently neglects inherent inter-view heterogeneity, thereby risking representation homogenization across views. While masking mechanisms can alleviate these issues, the high masking ratio, which is necessary for generating discriminative representations in single-view scenarios, forces graph-based models to produce diverse view-specific representations at the expense of significant consistency loss, i.e., semantically misaligned representations. To address these challenges, this paper proposes Dual Contrastive Masked Graph-Autoencoder Learning (DCMGAL) for multi-view clustering. DCMGAL incorporates a masked aggregation module that amplifies inter-view discrepancies while suppressing noise and redundancy through randomized edge masking. It further designs a global feature fusion mechanism, which integrates dual attention networks with a self-expression network to capture complementary information across views. Moreover, a dual contrastive learning module is devised to enforce cluster-level consistency via cross-view cluster consistency learning and preserve local topological structure via local topology preservation learning, thereby enhancing representation separability. Finally, an adjacency graph reconstruction component is incorporated to retain essential neighborhood information via a graph autoencoder architecture, while simultaneously reconstructing node features. Extensive experiments on benchmark datasets show that DCMGAL significantly outperforms state-of-the-art clustering methods.

1. Introduction

Multi-view clustering (MVC) has attracted significant research attention in recent years, as it enables the analysis of multi-view data without reliance on labeled annotations. Unlike single-view data, which offer limited perspectives on data objects, multi-view data facilitate comprehensive characterization by integrating heterogeneous views, where each view captures distinct semantics while preserving complementary information. Collectively, these views reveal deep underlying data structures that are inaccessible to single-view approaches (Chao et al., 2021; Liang et al., 2020; Zhang et al., 2024). Consequently, recent advancements in MVC research are directed toward integrating cross-view complementary information. This integration aims to more effectively leverage the underlying data structures by jointly exploring both cross-view

consistency and view-specific diversity (Cai et al., 2024; Zhang & Che, 2024), especially for complex scenarios such as those with unbalanced or incomplete views.

Traditional MVC methodologies (Guo & Wu, 2025; Liang et al., 2020) primarily rely on shallow data representations, such as handcrafted features or linear embeddings derived from matrix factorization techniques. However, these shallow representations are often low-quality, containing redundancy and noise that limit their ability to model the complex nonlinear relationships inherent in multi-view data (Xu et al., 2021, 2023; Yan et al., 2025).

To address these limitations, recent studies have developed deep MVC frameworks (Cui et al., 2023; Xu et al., 2021) that leverage deep neural networks for nonlinear mapping. Among these approaches, the deep autoencoder architecture has emerged as a prevalent paradigm

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Table 1

Quantitative Evaluation of Major Issues in Multi-view Clustering (NS = Noise Sensitivity, VR = View Redundancy, HU = Heterogeneity Utilization).

Metric	Datasets	Baseline Methods	DCMGAL (Ours)
NS (%)	Caltech-2V	14.1 ± 3.5	7.9 ± 1.2
	BDGP	15.6 ± 1.9	8.3 ± 2.5
	Cora	20.7 ± 3.1	8.4 ± 1.8
VR	Caltech-2V	0.57 ± 0.08	0.34 ± 0.02
	BDGP	0.67 ± 0.02	0.36 ± 0.04
	Cora	0.51 ± 0.01	0.35 ± 0.03
HU (%)	Caltech-2V	28.9 ± 2.6	40.2 ± 3.1
	BDGP	29.4 ± 1.7	42.8 ± 3.6
	Cora	24.6 ± 2.4	41.5 ± 2.9

Note: This table reports the best performance among six baseline methods (SiMVC, MFLVC, DealMVC, MRDD, DDMVC, and DCMVC) alongside the results of our proposed DCMGAL on the Caltech-2V, BDGP, and Cora datasets.

(Xu et al., 2021; Yan et al., 2023) in MVC, using encoder-decoder architectures to encode input data into informative low-dimensional representations while retaining critical information via data reconstruction. Nonetheless, a key limitation of these methods lies in their overemphasis on feature-level reconstruction, often neglecting topological structures that encode critical pairwise sample relationships.

Graph-based deep MVC methods (Peng et al., 2019; Wang et al., 2020; Wen et al., 2021) address this gap by integrating feature and structural information via graph convolutional networks (GCNs). These methods aggregate sample features with neighborhood information through adjacency-guided propagation to generate neighborhood-consistent representations that adhere to neighborhood smoothness assumptions. Despite these advancements, three critical challenges persist: (1) **Sensitivity to redundancy and noise**: The presence of redundancy and noise in data results in graph structures with redundant and noisy connections, which GCNs tend to overfit, thereby compromising model performance and reliability (Peng et al., 2024). (2) **Insufficient diversity and distinguishability of representations**: Existing approaches pursue cross-view semantics but overlook inter-view inconsistency, risking representation homogenization (Cai et al., 2024; Zhang & Che, 2024). In such cases, view-specific representations become overly similar, losing critical diversity necessary for capturing view-specific semantics. (3) **Imbalance between feature and structural information**: While GCNs can concurrently model feature and structural information, existing models overemphasize feature reconstruction but neglect structural reconstruction. Consequently, generated representations may reconstruct data effectively but fail to retain neighborhood information, critical for characterizing the manifold structure of multi-view data.

Table 1 presents a comparative analysis of **Noise Sensitivity (NS)** (Shen & Kang, 2025), **View Redundancy (VR)** (Shen et al., 2024), and **Heterogeneity Utilization (HU)** across six state-of-the-art multi-view clustering methods and our proposed DCMGAL. The benchmark methods include SiMVC (Trosten et al., 2021), MFLVC (Xu et al., 2022), DealMVC (Yang et al., 2023), MRDD (Ke et al., 2024), DDMVC (Xu et al., 2025), and DCMVC (Li et al., 2025). **Noise Sensitivity (NS)** quantifies the robustness of a model to structural perturbations. It is defined as the relative percentage decrease in clustering Accuracy (ACC) after injecting 20% random noise into the adjacency matrix of each view. A lower NS value indicates greater resilience to structural noise. **View Redundancy (VR)** measures the degree of information overlap across views, defined as the average cosine similarity between representations of different views. A higher VR indicates more redundant information shared between views. **Heterogeneity Utilization (HU)** evaluates the capacity of a method to leverage view-specific features. It is defined as the relative drop in Normalized Mutual Information (NMI) after ablating view-specific features (i.e., retaining only

shared features across views). The view-specific features are isolated by computing the residual between the original view features and their projection onto the shared subspace. A lower HU reflects a weaker ability to exploit heterogeneous information. The quantitative results in Table 1 explicitly validate the limitations of current MVC methods regarding sensitivity to noise, redundancy, and under-utilization of heterogeneous information. Concurrently, the results demonstrate the superior robustness, efficient leveraging of complementary information, and strong capacity to exploit view-specific features inherent to the proposed DCMGAL framework.

To tackle these challenges, this paper proposes a novel deep MVC framework termed Dual Contrastive Masked Graph-Autoencoder Learning (DCMGAL). DCMGAL incorporates a **masked graph-based aggregation module** that randomly masks edges in adjacency graphs. This module introduces inter-view inconsistency, mitigating noise propagation and redundant connections while enhancing robustness in view-specific representation learning. However, this manually induced inconsistency presents a trade-off: high masking ratios (required to boost representation learning capacity (He et al., 2022; Tan et al., 2023)) risk severing critical edges that encode cross-view consensus structures, disrupting the neighborhood smoothness assumption underlying graph convolutional propagation.

This disruption weakens the ability of the model to capture inter-view statistical dependencies and fragments latent manifolds, rendering semantically aligned cross-view samples topologically disconnected. Consequently, representations learned under high masking ratios exhibit increased heterogeneity and reduced consistency, leading to poor discriminability and unstable clustering performance. Accordingly, a **global fusion mechanism** is designed to restore cross-view semantic alignment and reinforce latent structural correlations. It utilizes dual attention networks to dynamically model inter-view correlations and emphasize informative connections, followed by a self-expression network to reveal intrinsic data structures, effectively capturing complementary information. Ablation studies and t-SNE visualizations in Section 4.5 validate that omitting either submodule leads to overly dispersed and semantically misaligned representations under high masking ratios (e.g., $\omega \geq 50\%$), significantly degrading cluster separability. Additionally, DCMGAL devises a **dual contrastive learning module** to enhance the discriminability of fused representations, consisting of two key components: cluster-level consistency learning and topology-level structural preservation learning. By enforcing clustering consistency across views while retaining local topology structures, this module ensures that similar samples are brought closer together, while dissimilar ones are separated in the feature space. Finally, DCMGAL jointly reconstructs input features and view-specific neighborhood structures to enhance the neighborhood smoothness of latent representations. Fig. 1 illustrates the framework of DCMGAL. The primary contributions are summarized as follows:

- A masked graph-based aggregation module is developed to suppress noise propagation and redundant connections in adjacency graphs, amplify inter-view heterogeneity, and improve robustness in graph-based representation learning.
- A global feature fusion mechanism is designed to restore cross-view semantic alignment under high masking ratios via attention-driven correlation exploration and self-expressive latent structure learning.
- A dual contrastive learning module is devised to enforce cluster-level cross-view consistency and topology-level structural preservation, jointly enhancing representation discriminability.

2. Related work

Deep multi-view clustering research currently centers on deep autoencoder-based methodologies. Representative studies (Trosten et al., 2021; Xu et al., 2022; Yan et al., 2023; Yang et al., 2023) employ deep autoencoders to generate view-specific latent representations,

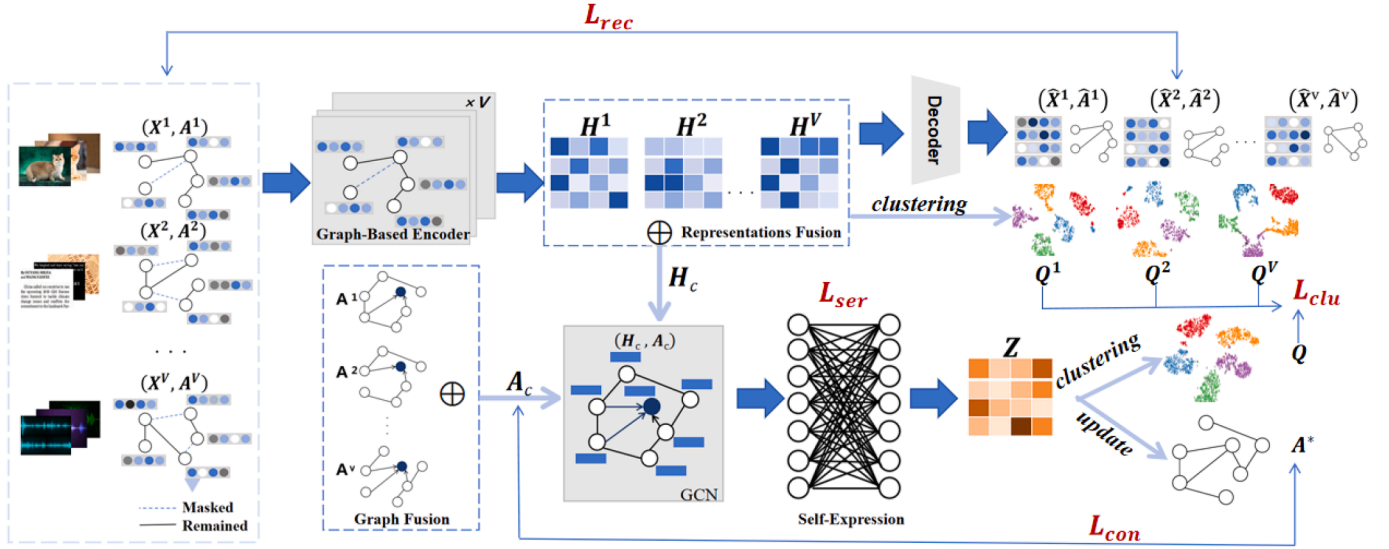


Fig. 1. Framework of the proposed DCMGAL. The proposed framework contains three core components: a masked aggregation module, a global fusion module and a dual contrastive learning module. The masked aggregation module initiates the process by generating view-specific representations via masked GCN-based autoencoders. To capture inter-view complementary information, the framework then integrates these view-specific representations via the global fusion module, which combines self-attention and self-expression networks. Finally, the dual contrastive learning module facilitates the extraction of clustering-friendly representations that preserve neighborhood consistency across views.

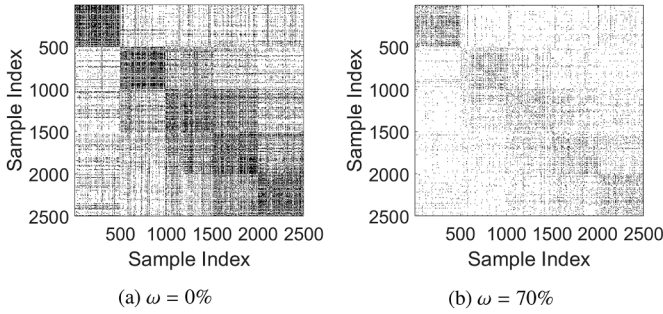


Fig. 2. Element-wise product of view-specific adjacency matrices on BDGP when the masked ratios are 0% and 70%.

with cross-view alignment enforced via hierarchical feature matching. Typically, these methods utilize attention mechanisms to dynamically weight cross-view correlations, prioritizing semantically salient features to enhance representation fusion. To further improve representation separability, contrastive learning frameworks have been extensively adopted (Xu et al., 2023; Yan et al., 2025; Zhang et al., 2026). These frameworks maximize mutual information between samples of positive pairs while enforcing divergence between samples of negative pairs. Despite their efficacy, a critical limitation of such methods is their predominant focus on instance-level discrimination, often overlooking neighborhood information essential for capturing intrinsic data manifold structures.

With the advancement of deep learning in the field of graph representation learning, numerous graph-based deep multi-view clustering approaches have been proposed (Wang et al., 2020; Wen et al., 2021). These methods leverage graph convolutional networks to aggregate neighborhood information through adjacency graphs, thereby encoding local geometric structures. Specifically, they construct view-specific adjacency graphs to model local neighborhoods while employing attention mechanisms to explore cross-view consistency. Notably, despite achieving satisfactory feature reconstruction, their generated representations remain suboptimal for clustering tasks. To obtain clustering-friendly representations, self-supervised mechanisms (Bo et al., 2020; Huang et al., 2023; Peng et al., 2019) refine clustering probability distributions

during model training. Nevertheless, these methods remain susceptible to noise propagation and structural biases in graph construction, especially when handling redundant or noisy multi-view data.

Masked learning strategies have exhibited remarkable efficacy in reducing redundancy and enhancing representation robustness via reconstructing masked content from contextual information. This paradigm involves random masking of input features or structures, followed by prediction of the masked elements, which has yielded promising performance gains across diverse domains (He et al., 2022; Hou et al., 2022; Tan et al., 2023). To advance vision learners, masked learning strategies have been extensively adopted in computer vision (Bachmann et al., 2022; Bao et al., 2022; Gong et al., 2023; Huang et al., 2024; Jamal & Mohareri, 2025). Partial masked vision learners (Bao et al., 2022; Huang et al., 2024) construct bi-view image data and mask only the image patches of the original data. They feed unmasked image patches into an encoder and utilize the unmasked counterpart to supervise masked data reconstruction. In contrast, fully masked vision learners (Bachmann et al., 2022; Gong et al., 2023; Jamal & Mohareri, 2025) apply masking to all views of multi-modal data. They explore cross-modal relationships for masked patch reconstruction via a shared encoder or contrastive learning on unmasked image patches. However, these methods are especially designed for vision data. The recent work (Ke et al., 2024) pioneers masked modeling for heterogeneous multi-view data via random feature masking, using a two-stage learning framework: Stage 1 learns shared representations with a shared encoder and reconstructs masked features using view-specific decoders based on these shared representations; Stage 2 learns view-specific information from raw data, combining the shared representations of Stage 1 to reconstruct the original data.

However, these methods exhibit distinct limitations. First, while they aim to enhance their ability to learn shared representations and model cross-view relationships through data recovery, shared encoders, and contrastive learning, they fail to address semantic misalignment induced by high masking ratios. They indiscriminately drop features, yet provide no mechanism to address the resulting fragmentation, undermining both clustering stability and representation distinguishability. Second, existing methods overlook critical graph structure information, yielding topology-unaware representations that fail to exploit neighborhood relationships.

To overcome these limitations, we propose DCMGAL, a dual contrastive masked graph-autoencoder learning method for multi-view clustering. Our framework applies edge masking on adjacency graphs to explicitly induce inter-view heterogeneity while mitigating noise, coupled with GCN autoencoders that capture view-specific feature-topology representations. To further integrate complementary cues, a global fusion mechanism combining an attention-guided fusion module and a self-expression network is introduced. Moreover, a dual contrastive learning mechanism is designed to enforce cluster-level consistency across views while preserving local topological structures, ensuring that the learned representations are both discriminative and structure-aware. Extensive ablation studies confirm the individual contribution of each component, underscoring the effectiveness and necessity of the proposed modules.

3. Methodology

3.1. The framework of DCMGAL

As illustrated in Fig. 1, the proposed DCMGAL framework integrates three key components: (1) a masked graph-based aggregation module for robust view-specific representation learning, (2) a global fusion mechanism to restore cross-view semantic consistency, and (3) a dual contrastive learning module to enhance representation discriminability. Initially, view-specific GCN encoders with randomized edge masking are developed to generate robust and heterogeneous representations by suppressing noise and redundancy. These view-specific representations are subsequently fused via a dual attention mechanism and a self-expression network to capture complementary information and uncover intrinsic data correlations across views, mitigating semantical misalignment induced by edge masking. Finally, a dual contrastive learning module is formulated to further refine the representations, yielding clustering-friendly and neighborhood-consistent embeddings.

3.2. Problem formulation

Given multi-view data $\mathcal{X} = \{X^{(1)}, X^{(2)}, \dots, X^{(V)}\}$ with N samples across V views, where $X^v \in \mathbb{R}^{N \times d_0^v}$ denotes the data matrix of the v -th view, DCMGAL aims to learn discriminative representations $Z \in \mathbb{R}^{N \times m}$ and partition them into K semantically coherent clusters.

3.2.1. Masked graph-based aggregation

Representation Learning via GCN-Based Autoencoders

To exploit view-specific feature-structure relationships, a symmetric GCN encoder-decoder architecture is established for each view. These GCN encoders transform input data $\mathcal{X}$ into view-specific latent representations $H^v \in \mathbb{R}^{N \times m}$ through neighborhood aggregation, while GCN decoders reconstruct both input features and neighborhood structures to preserve manifold information.

The key operation of GCN involves updating node representations by aggregating feature information from nodes and their neighbors. Typically, the neighborhood structure is represented by an adjacency matrix for each data view. Let A^v denote the adjacency matrix of the v -th view. The representations $H^{(v,l)}$ at the l -th layer of the encoder are then computed as:

$$H^{(v,l)} = \text{ReLU}\left((D^v)^{-\frac{1}{2}} \tilde{A}^v (D^v)^{-\frac{1}{2}} H^{(v,l-1)} W^{(v,l)}\right), \quad (1)$$

where $\tilde{A}^v = A^v + I$ denotes the adjacency matrix with self-connections, D^v is the degree matrix of A^v with $D_{ii}^v = \sum_j A_{ij}^v$, $W^{(v,l)} \in \mathbb{R}^{d_{l-1}^v \times d_l^v}$ is a trainable projection matrix, and $H^{(v,0)} = X^v$.

For data reconstruction, the decoder network of each view adopts an architecture symmetric to the encoder. The final output of each decoder is the reconstructed data matrix $\hat{X}^v$.

Consequently, the quality of the learned representations H^v is initially assessed via the reconstruction loss:

$$L_{rec} = \sum_{v=1}^V \|X^v - \hat{X}^v\|_F^2. \quad (2)$$

However, this formulation exclusively reconstructs feature information while neglecting structural information recovery, risking loss of neighborhood information critical to model the manifold structure inherent in multi-view data. Thus, the reconstruction loss is extended to include adjacency matrix reconstruction, ensuring the learned representations maintain view-specific neighborhood structures:

$$L_{rec} = \sum_{v=1}^V \|X^v - \hat{X}^v\|_F^2 + \sum_{v=1}^V \|A^v - \hat{A}^v\|_F^2, \quad (3)$$

where $\hat{A}^v$ denotes the adjacency matrix derived from $\hat{X}^v$. Specifically, the reconstructed adjacency matrix $\hat{A}^v$ is generated via a differentiable relaxation of the KNN-based strategy (with $k = 10$ and cosine similarity metric) to address the non-differentiability inherent in hard KNN graph construction. To circumvent this non-differentiability and enable end-to-end gradient flow, the mapping function $\hat{A}^v = f(\hat{X}^v)$ is defined as a soft probabilistic assignment:

$$\hat{A}_{ij}^v = \frac{\exp(\cos(\hat{x}_i^v, \hat{x}_j^v)/\tau)}{\sum_{k \neq i} \exp(\cos(\hat{x}_i^v, \hat{x}_k^v)/\tau)}, \quad (4)$$

where $\tau = 0.1$ is a temperature parameter controlling the smoothness of the similarity distribution, and $\hat{x}_i^v$ denotes the i -th sample in $\hat{X}^v$. This softmax-based relaxation effectively replaces the hard binary assignment of traditional KNN graphs with a continuous and fully differentiable operation. The cosine similarity between reconstructed features $\cos(\hat{x}_i^v, \hat{x}_j^v)$ is inherently differentiable with respect to $\hat{X}^v$, and the subsequent softmax normalization preserves this differentiability. During the inference phase, the soft similarity scores are thresholded (setting $\hat{A}_{ij}^v = 1$ if $\hat{A}_{ij}^v > 0.5$, and 0 otherwise) to recover a binary adjacency matrix, thereby ensuring consistency with the original graph A^v .

Notably, all adjacency graphs in this work are constructed using the k -nearest neighbor method with the cosine similarity metric, where k is set to 10.

Masked Adjacency Graphs

GCN-based encoders are inherently susceptible to redundancy and noise in multi-view data, leading to overly homogeneous representations that lack inter-view diversity.

To address these issues, DCMGAL implements randomized edge masking on view-specific adjacency graphs. For each view, edges are randomly removed with a predefined masking ratio ω , and corresponding entries in the adjacency matrix A^v are set to zero. This masking mechanism effectively enhances inter-view heterogeneity and reduces redundancy. As empirically demonstrated in Fig. 2, cross-view structural patterns are significantly reduced under a 70% masking ratio, facilitating the learning of diverse and robust view-specific representations.

3.2.2. Global fusion mechanism

Aggressive edge masking (e.g., $\omega = 70\%$) effectively amplifies inter-view heterogeneity and suppresses noise propagation but concurrently disrupts critical edges that encode consensus structure across views. This disruption violates the neighborhood-smoothness prior inherent to GCNs and fragments the latent manifold: semantically aligned samples across views become topologically disconnected, yielding representations with high heterogeneity yet low discriminability. Ablation studies and t-SNE visualizations in Section 4.5 demonstrate that high masking ratios risk degrading representation quality and inducing semantical misalignment. However, high masking ratios are necessary to enhance representation robustness and discriminability (He et al., 2022; Tan et al., 2023). To counteract this trade-off, the proposed DCMGAL integrates a global fusion mechanism, comprising an attention-based fusion submodule to recover informative inter-view connections and a self-expression submodule to rebuild an intact structure.

Dual Attention Fusion

Following the derivation of view-specific representations, DCMGAL computes consensus representations H_c and an adjacency matrix A_c via two attention fusion networks.

Specifically, a feature attention fusion network is formulated to fuse view-specific representations H^v as follows:

$$H_c = \sum_{v=1}^V w_h^v [H^v, \mathbf{1}] \Theta_c, \quad (5)$$

where $\Theta_c \in \mathbb{R}^{(m+1) \times m}$ is a trainable attention matrix, and the view weights w_h^v are computed via softmax-normalized attention coefficients:

$$w_h^v = e^{\alpha_v} / \sum_{p=1}^V e^{\alpha_p}. \quad (6)$$

The attention coefficient α_v is defined as $\alpha_v = \tanh([\mathbf{h}_c^v, \mathbf{1}]^T \mathbf{q}_v)$ (Wu et al., 2019), where $\mathbf{q}_v \in \mathbb{R}^{1 \times m}$ and $\mathbf{K}^v \in \mathbb{R}^{(m+1) \times m}$ are the query vector and key matrix, respectively.

Similarly, the consensus adjacency matrix A_c is derived by fusing view-specific adjacency matrices A^v using a structural attention fusion network:

$$A_c = \sum_{v=1}^V w_a^v A^v, \quad (7)$$

where w_a^v denotes the weight assigned to the adjacency matrix A^v . The view weight w_a^v for the v -th view is determined by the degree of node aggregation β_v (Zhao et al., 2024):

$$\beta_v = \sum_{i=1}^n \sum_{j=1}^n (-l_{ij} \ln S_{ij}^v - (1 - l_{ij}) \ln(1 - S_{ij}^v)), \quad (8)$$

$$w_a^v = e^{\beta_v} / \sum_{i=1}^V e^{\beta_i}, \quad (9)$$

where $S_{ij}^v = \cos(x_i^v, x_j^v)$ denotes the cosine similarity between x_i^v and x_j^v , and $l_{ij} = 1$ if nodes x_i and x_j belong to the same cluster, and 0 otherwise. Due to the unsupervised nature of multi-view clustering, ground-truth class labels are not available. Consequently, l_{ij} is dynamically estimated during training by leveraging the progressively refined feature representations to capture the evolving data structure. The estimation of l_{ij} follows a two-phase strategy. In the initialization phase, l_{ij} is initialized using the connectivity of the KNN graph with $k = 10$. Specifically, $l_{ij} = 1$ if samples x_i and x_j are neighbors, and $l_{ij} = 0$ otherwise. This provides a principled structural prior based on the local manifold geometry. As training progresses and the feature representations are refined, l_{ij} is dynamically updated based on the intermediate cluster assignments. This enables the relational guidance to adapt to the increasingly discriminative latent data structure. Importantly, l_{ij} is used exclusively within the structural attention fusion module to compute the view weight w_a^v .

Finally, the consensus representations H_c and consensus adjacency graph A_c are processed through a single-layer GCN to refine H_c :

$$Y = \text{ReLU}(D_c^{-\frac{1}{2}} \tilde{A}_c D_c^{-\frac{1}{2}} H_c W_c), \quad (10)$$

where $\tilde{A}_c = A_c + I$ and D_c is the corresponding degree matrix.

Self-Expression Reconstruction

Data typically lie in a union of low-dimensional subspaces, indicating that each sample can be effectively represented as a linear combination of other samples within the same subspace (Zhang et al., 2019). By exploring self-expression correlations between data, the model automatically suppresses redundant and irrelevant features while preserving those critical for clustering. Consequently, after deriving consensus representations via a single-layer GCN, a fully connected self-expression network is introduced to explore the underlying subspace structure by minimizing:

$$L_{ser} = \lambda_0 \|C\|_1 + \frac{1}{2} \|Y - YC\|_F^2, \text{ s.t. } \text{diag}(C) = 0, \quad (11)$$

where the constraint $\text{diag}(C) = 0$ prevents the trivial solution $C = I$ (Ji et al., 2017), and $\lambda_0 > 0$ is a trade-off parameter. The l_1 -norm $\|C\|_1$ enforces sparse linear combinations, ensuring that each sample is expressed by a small set of neighbors and globally exploring the neighborhood structure of multi-view data. Consequently, the final consensus representations Z are derived as $Z = YC$.

Algorithm 1 Dual Contrastive Masked Graph-Autoencoder Learning (DCMGAL).

- 1: **Input:** Multi-view data $X = \{X^{(1)}, X^{(2)}, \dots, X^{(V)}\}$; Number of clusters K ; Masking ratio ω ; Regularization parameters λ_0, λ_1 , and λ_2 ; Number of epochs T .
- 2: **Output:** Clustering result $\mathcal{Q}$.
- 3: **Initialization:**
Mask adjacency matrix $A^{(v)}$ under the masking ratio of ω .
Pre-train deep autoencoders and initialize model parameters by optimizing Eq. (3).
- 4: **while** not reaching T **do**
- 5: Obtain view-specific representations $H^{(v)}$ via Eq. (1).
- 6: Refine consensus representations Z using attention mechanism and self-expression network (Eq. (10) and Eq. (11)).
- 7: Calculate multi-view soft clustering assignment $Q^{(v)}$ via Eq. (12).
- 8: Perform dual contrastive learning by optimizing Eq. (14) and Eq. (15).
- 9: Fine-tune DCMGAL model parameters by optimizing Eq. (16).
- 10: **end while**
- 11: Obtain clustering result C by k -means + +.

3.2.3. Dual contrastive learning

While consensus representations derived from view-specific representations effectively integrate complementary information, they frequently lack sufficient discriminability for clustering tasks. To mitigate this shortcoming, a dual contrastive learning module is included, comprising cross-view clustering consistency learning and local topology preservation learning. This module enhances representation separability by enforcing cluster-level alignment across views and retaining the underlying data manifold structure, thereby generating clustering-friendly and topology-aware representations.

Cross-view Clustering Consistency Learning

The component is formulated to achieve clustering consistency and generate clustering-friendly representations by aligning view-specific cluster structures with the consensus structure. For notational simplicity, consensus representations Z are denoted as H^{V+1} in the following.

First, k -means + + (Sculley, 2010) is employed to derive cluster assignments for each view (including the consensus view H^{V+1}). Let μ_c^v denote the c -th cluster center of the v -th view. Then, the soft clustering assignment Q_{ic}^v is computed as:

$$Q_{ic}^v = \frac{(1 + \|\mathbf{h}_i^v - \mu_c^v\|^2)^{-1}}{\sum_l (1 + \|\mathbf{h}_i^v - \mu_l^v\|^2)^{-1}}. \quad (12)$$

Subsequently, DCMGAL ensures that soft clustering assignments of the same cluster across views are consistent, while those of different clusters are mutually exclusive. Formally, positive and negative pairs are constructed as (q_c^v, q_c^m) and (q_c^v, q_l^m) , respectively. Then, the contrastive loss between the v -th and m -th view is given by:

$$L^{vm} = -\frac{1}{K} \sum_{c=1}^K \log \frac{e^{G_{cc}^{vm}/\tau}}{\sum_{l \neq c}^K e^{G_{cl}^{vm}/\tau} + \sum_{l=1}^K e^{G_{cl}^{vm}/\tau}}, \quad (13)$$

where $G_{cl}^{vm} = \cos(q_c^v, q_l^m)$ and τ is the temperature parameter.

By aggregating pairwise view losses L^{vm} , the total clustering consistency loss is derived as:

$$L_{clu} = \frac{1}{2} \sum_{v=1}^{V+1} \sum_{m \neq v}^{V+1} L^{vm} + \sum_{v=1}^{V+1} \sum_{c=1}^K f_c^v \log f_c^v. \quad (14)$$

where $f_c^v = \frac{1}{N} \sum_{i=1}^N Q_{ic}^v$. Notably, $\sum_{v=1}^{V+1} \sum_{c=1}^K f_c^v \log f_c^v$ is incorporated to mitigate cluster imbalance, wherein certain clusters may otherwise be excessively small or empty.

Local Topology Preservation Learning

The effectiveness of cross-view clustering consistency learning inherently depends on the quality of consensus representations. To improve their separability, a local topology preservation component is

developed, as local topological structures encode the underlying data manifold, and neighborhood-consistent representations greatly improve clustering performance. To this end, positive and negative pairs are constructed as follows: (1) If two nodes are neighbors in both the consensus graph $\mathcal{G}_c$ (with adjacency matrix A_c) and the target graph $\mathcal{G}^*$ (with adjacency matrix A^* that is derived from representations Z), indicating strong semantic similarity, they form a positive pair; and (2) If two nodes are neighbors in either $\mathcal{G}_c$ or $\mathcal{G}^*$, indicating weak similarity and thus classified as “fake neighbors” (ostensibly neighboring yet insufficiently similar), they form a negative pair. This design enforces proximity between nearest neighbors in the latent space while promoting distance between fake neighbors, thus generating well-separated representations.

Denote Z_i^+ and Z_i^- as the sets of consensus representations corresponding to positive and negative samples of the i -th sample x_i , respectively. The local topology preservation loss is then defined to maximize correlations between samples of positive pairs while minimizing correlations between samples of negative pairs:

$$L_{con} = - \sum_{i=1}^N \log \frac{\text{Corr}(z_i, Z_i^+)}{\text{Corr}(z_i, Z_i^+) + \text{Corr}(z_i, Z_i^-)}, \quad (15)$$

where $\text{Corr}(z, Z) = \sum_{z_j \in Z} e^{\cos(z, z_j)/\tau}$.

By integrating Eqs. (3), (11), (14), and (15), the total loss function of DCMGAL is formulated as:

$$L = L_{rec} + L_{ser} + \lambda_1 L_{clu} + \lambda_2 L_{con}, \quad (16)$$

where λ_1 and λ_2 are parameters balancing the contributions of contrastive losses.

The training algorithm is outlined in Algorithm 1.

3.3. Discussion

3.3.1. Complexity analysis

This section provides a detailed analysis of the computational complexity of the proposed DCMGAL for multi-view data comprising N samples, V views, and an average feature dimension of d per view, by decomposing the costs of its major components. The analysis focuses on the per-epoch training cost, excluding the one-time graph construction preprocessing.

The masked graph-based aggregation module, which employs three-layer GCN encoders and decoders for each view, incurs a cost dominated by the neighborhood aggregation operations. Given the use of KNN graphs with $k = 10$, the adjacency matrix A^v for each view contains approximately $10N$ non-zero entries. Consequently, the complexity for a single GCN layer is $O(|E^v| \cdot d) = O(10Nd)$. With V views and $L = 3$ layers, the total complexity for this module scales as $O(V \cdot L \cdot 10Nd) = O(30VNd)$.

The global fusion mechanism integrates two components: a dual attention fusion module and a self-expression network. The dual attention fusion, comprising feature and structural attention modules, performs matrix multiplications with dimensions $N \times m$ (where m is the consensus representation dimension) and $N \times N$, resulting in a complexity of $O(VNm + VN^2)$. In contrast, the self-expression network constitutes the primary computational bottleneck. Its core operation involves multiplying the consensus representation matrix $Y \in \mathbb{R}^{N \times m}$ by the coefficient matrix $C \in \mathbb{R}^{N \times N}$, resulting in a quadratic complexity of $O(N^2m)$.

The dual contrastive learning module encompasses two objectives: cross-view cluster consistency and local topology preservation. The former calculates cosine similarities between cluster centers across $V + 1$ views (including the consensus view) for K clusters, with a complexity of $O((V + 1)^2 K^2)$. The latter constructs positive and negative sample pairs based on the consensus graph A_c and the target graph A^* . Both graphs are KNN-based with roughly $10N$ edges each, leading to a combined complexity of $O(|E_c| + |E^*|) \approx O(20N)$. Therefore, the overall complexity for this module is $O((V + 1)^2 K^2 + N)$.

Additional costs arise from the reconstruction and loss calculation components. These involve Frobenius norm computations for feature and adjacency matrices, resulting in a complexity of $O(VNd + VN^2)$.

Synthesizing the costs from all components, the total per-epoch computational complexity of DCMGAL is: $O(VLNd + N^2m + VN^2 + (V + 1)^2 K^2)$. For large-scale datasets where the number of samples N significantly exceeds other parameters (d , V , and K), the complexity is dominated by the quadratic term $O(N^2m)$, which originates from the self-expression network. This quadratic scaling is consistent with most subspace clustering methodologies. A comparative analysis with state-of-the-art baseline methods shows that the complexity of DCMGAL is comparable: it aligns with the $O(VN^2 + Nd)$ complexity of GCFAgg and the $O(N^2 + VNd)$ complexity of DCMVC.

Empirical validation on the Fashion dataset ($N = 10^4$) confirms the practical efficiency of the framework. Utilizing a single NVIDIA Tesla T4 GPU, DCMGAL completes one training epoch in approximately 12.3 seconds. This runtime is only 1.2 times that of GCFAgg (10.1 seconds) and 1.1 times that of DCMVC (11.2 seconds), demonstrating a computationally viable profile for medium-to-large-scale datasets.

In summary, the DCMGAL framework exhibits a per-epoch computational complexity of $O(N^2)$, which is comparable to contemporary state-of-the-art multi-view clustering methods. However, the self-expression module remains a scalability bottleneck for very large datasets. Future work will therefore concentrate on developing computationally efficient approximations, such as low-rank or sparse representations, to reduce this quadratic complexity without compromising clustering performance.

3.3.2. Rationale and limitations of the fixed masking ratio of 70%

The setting of a fixed 70% masking rate is a calibrated design choice, balancing the suppression of noise and redundancy, the preservation of cross-view semantics, and computational efficiency. This choice is grounded in empirical evidence from recent advances in mask-based multi-view learning and is specifically adapted to the two-stage distillation-disentanglement architecture of our framework.

A masking ratio of 70% lies within the empirically effective range (60%-75%) identified for tasks requiring denoising (Jamal & Mohareri, 2025; Ke et al., 2024). This rationale is validated across four representative datasets with diverse feature types (see Subsection 4.4 for details): Caltech-2V/5V (images) (Fei-Fei et al., 2004), BDGP (biology) (Cai et al., 2012), Cora (citations) (Fang et al., 2024), and MNIST (digits) (Peng et al., 2019). The validation reveals a common challenge: adjacency graphs constructed from raw data features contain a substantial proportion of edges that are either redundant (intra-class but non-semantic connections) or noisy (false inter-class connections). For instance, in Caltech-2V/5V, SIFT-based adjacency graphs contain substantial redundant edges (intra-class non-key texture connections) and noisy edges (false connections caused by background interference); in BDGP, gene expression view graphs suffer from redundant connections due to functional co-expression and measurement errors; in Cora, the citation graph is affected by redundant mutual citations among topically overlapping papers and false citations to weakly relevant cross-disciplinary work; and the pixel-similarity graphs contain redundant intra-digit non-shape connections and noisy inter-digit false connections arising from similar pixel distributions across different digits in MNIST. The 70% masking ratio effectively eliminates the majority of these non-essential edges while retaining a sparse backbone of core edges that encode fundamental semantic relationships (e.g., object shape similarity, key gene regulatory links, core citation chains, and digit contours). This is possible because the core semantic edges themselves exhibit redundancy in the raw graphs. This targeted pruning directly addresses the critical challenge in graph-based multi-view clustering: the misleading effect of noisy and redundant edges on representation learning.

Although the heavy masking risks semantic corruption, our framework, DCMGAL, provides robust compensation for potential semantic loss by integrating dual contrastive learning and attention-based

consensus fusion. It establishes a synergistic “masking-pruning and semantic compensation” mechanism. Specifically, by constructing positive pairs from the masked and unmasked representations of the same sample and negative pairs from different samples, the model is guided to discard view-specific noise and focus on intrinsic semantic features, effectively mitigating the impact of edge pruning. For example, this process enables the model to prioritize digit contours in MNIST over noisy pixel correlations, core topic citations in Cora over fake cross-domain links, object shape similarity in Caltech-2V/5V over background texture interference, and key gene regulatory patterns in BDGP over random co-expression signals. Additionally, the attention-based fusion module dynamically weights view-specific representations based on their structural integrity after masking. For instance, in BDGP, the gene sequence view (which contains fewer noisy edges) receives a higher weight; whereas the image shape view (which retains more core edges) dominates consensus fusion in Caltech-2V/5V. This ensures that views with higher semantic reliability contribute more substantially to the final consensus representation Z . This compensation mechanism recovers the majority of semantic information lost due to masking, as evidenced by the consistency between Z and the ground-truth semantic labels.

Although a fixed masking ratio of 70% achieves stable performance on most datasets (consistent with the established empirical setting (Ke et al., 2024)), it exhibits limitations in adapting to heterogeneous multi-view data. Different views within the same dataset, as well as different datasets, often exhibit distinct levels of noise and redundancy. For example, the noise level in the image shape view of Caltech-2V/5V is lower than that in the texture views; the gene sequence view of BDGP contains fewer artifacts than its gene expression view. A single fixed ratio cannot accommodate such view-specific and dataset-specific variations. This inflexibility may result in insufficient masking for high-noise views or excessive masking for low-noise views, thereby hindering the generalization of DCMGAL to diverse multi-view data.

Adaptive masking strategies, such as view-adaptive and uncertainty-aware masking, are promising alternatives. However, implementing a robust adaptive strategy within the proposed DCMGAL introduces significant complexity and inherent challenges. Specifically, view-adaptive masking dynamically adjusts the masking ratio per view based on inherent redundancy (e.g., assigning higher ratios to noisy views), while uncertainty-aware masking infers edge reliability via model prediction uncertainty to mask high-uncertainty noisy edges. The primary challenges are twofold. First, the masking ratio is a hyperparameter with no learnable criterion to integrate it into the learning process of the model. Consequently, its adjustment lacks guidance from the training dynamics, making dynamical optimization based on data characteristics difficult. Second, these strategies lack an effective means to evaluate or guarantee generalization. Without a universally applicable validation standard, their performance may vary drastically across different dataset types (e.g., image vs. gene data) or task scenarios, failing to maintain stable effectiveness in multi-view clustering. Therefore, a fixed and empirically validated 70% ratio is employed to ensure stable and comparable performance across all benchmarks, maintaining a reproducible baseline for future research. The exploration of robust adaptive masking mechanisms remains a valuable direction for future work.

3.3.3. Extensibility to incomplete multi-view data

In real-world applications, incomplete multi-view data, where partial views of samples are missing due to sensor failures, data acquisition limitations, or experimental errors, poses a significant challenge to existing multi-view clustering methods, including the proposed DCMGAL. However, the robust learning and inference capabilities inherent in the masked graph aggregation module grant DCMGAL considerable potential for extension to such incomplete data scenarios. Specifically, by artificially masking 70% of edge connections during training, this module learns to capture intrinsic semantic correlations and robust invariant representations, which enables the model to infer meaningful semantic information for missing data from the available data in other views.

However, directly applying the original DCMGAL framework to incomplete multi-view scenarios faces significant challenges. The primary issue is the distortion of KNN graph construction, which occurs because the absence of data in missing views prevents accurate calculation of pairwise sample similarities, leading to incomplete or biased adjacency graphs. Two potential handling strategies are discussed below.

The first strategy is a straightforward three-step approach. First, the model is pre-trained on complete multi-view data to learn cross-view complementary information and establish global semantic consistency via attention-based global fusion and dual contrastive learning. Next, for missing data, their connections with other data are inferred based on the relationships observed in other views. Finally, guided by the preserved global semantic consistency, the model infers representations for the missing views and performs clustering. This imputation-based strategy is effective under low-to-moderate missing rates but becomes inadequate when the missing rate is high.

The second strategy constitutes a more complex approach, requiring specific modifications to the DCMGAL framework to handle high missing-rate scenarios effectively. This adaptation focuses on two key enhancements. First, the process of transferring KNN connections from available views to impute missing data should be made adaptive, based on a reliability measure for each data point derived from its completeness and cross-view consistency. This adaptation reduces the impact of missing data on graph construction. Second, the cross-view fusion module is enhanced by introducing a view-specific uncertainty estimation mechanism. This mechanism more accurately propagates semantic information from available data in other views to guide the inference of missing view representations, ensuring they remain consistent with the global semantic distribution.

3.3.4. Future research directions

Based on the above discussion regarding scalability, the masking mechanism, and extensibility to incomplete multi-view data, three key directions for future research are identified to further improve the performance, generalization, and applicability of the DCMGAL framework. First, the quadratic computational complexity poses a challenge for DCMGAL when applied to large-scale datasets. Hence, leveraging low-rank approximations to address this scalability bottleneck represents a promising future research direction. Second, developing a robust and interpretable adaptive masking strategy constitutes another critical research direction. This involves designing a view-adaptive and dataset-adaptive masking mechanism that moves beyond a fixed ratio to dynamically determine the optimal masking intensity for each view. Third, enhancing robustness for incomplete multi-view scenarios is a major priority. This would require a novel view-level attention fusion mechanism to dynamically weight the contributions of data views based on their reliability during cross-view semantic propagation. Additionally, integrating a view-specific uncertainty estimation module could enable adaptive adjustment of masking intensity on the preliminary adjacency graphs. Collectively, these enhancements aim to improve the retention of semantically core edges in extreme scenarios and thereby extend the applicability of the framework to complex real-world multi-view clustering tasks.

4. Experiments

4.1. Experimental setup

Datasets and Baselines

To evaluate the proposed DCMGAL model, experiments are conducted on seven publicly available multi-view datasets of varying scales across different domains, including **Caltech** (Fei-Fei et al., 2004), **100LEAVES** (Yang & Lin, 2023), **BDGP** (Cai et al., 2012), **Cora** (Fang et al., 2024), **MNIST** (Peng et al., 2019), **CCV** (Jiang et al., 2011), and **Fashion** (Xu et al., 2023). Table 2 provides a summary of the benchmark datasets.

Table 2
Description of the benchmark datasets.

Datasets	Views	Classes	Samples	Type
Caltech	5	7	1400	Image
100LEAVES	3	100	1600	Image
BDGP	2	5	2500	Gene
Cora	4	7	2708	Citation
MNIST	2	10	5000	Digit
CCV	3	20	6773	Image
Fashion	3	10	10,000	Image

Eleven state-of-the-art baseline methods are compared: **DEMVC** (Xu et al., 2021), **SIMVC** (Trosten et al., 2021), **CoMVC** (Trosten et al., 2021), **MFLVC** (Xu et al., 2022), **SDMVC** (Xu et al., 2023), **GCFagg** (Yan et al., 2023), **DealMVC** (Yang et al., 2023), **ACCMVC** (Yan et al., 2025), **MRDD** (Ke et al., 2024), **DDMVC** (Xu et al., 2025), and **DCMVC** (Li et al., 2025).

Implementation Details

The proposed DCMGAL is implemented using PyTorch. Each autoencoder comprises a three-layer GCN encoder and a symmetric decoder. Across all experiments, the hyper-parameters λ_0 , λ_1 , and λ_2 are configured as 0.1, 0.01, and 10, respectively, and the masked ratio is fixed at 70%. In the training process, the Adam optimizer is employed alongside a cosine annealing scheduler, with an initial learning rate of 3×10^{-4} and a batch size of 512. The masked graph-based aggregation module is pre-trained 2000 epochs, followed by 500 epochs of fine-tuning with the self-expression network frozen to retain the correlation structure, which is conducive to generating discriminative representations (He et al., 2005; Roweis & Saul, 2000). To ensure reliable convergence and prevent overfitting, we adopt an early stopping criterion. The criterion employs a patience parameter $P = 10$ and a loss tolerance $\epsilon = 10^{-4}$. Training is terminated if the absolute difference between the loss of the current epoch and the loss recorded P epochs earlier falls below ϵ , thereby eliminating unnecessary computational overhead once the convergence reaches a plateau. This early stopping technique serves as a common and effective form of implicit regularization, halting optimization when further updates yield negligible improvement and thereby balancing training efficiency with model performance. Preliminary analysis indicates that the DCMGAL framework typically attains peak performance within 1200 to 1600 epochs across the benchmark datasets. A maximum epoch limit ($T_{\max} = 2000$) is set to provide a sufficient buffer against underfitting. This maximum limit works in concert with the early stopping mechanism, which prevents overfitting from excessive training. To ensure fairness, all baselines are evaluated using official implementations and recommended settings.

Evaluation Metrics

The clustering performance is measured using three metrics, i.e., clustering accuracy (ACC), normalized mutual information (NMI), and purity (PUR) (Kumar et al., 2011).

4.2. Performance comparison

Table 3 presents clustering results on the Caltech dataset with 2–5 views. Optimal and suboptimal results are highlighted in **bold** and underline, respectively. The results demonstrate that: (1) DCMGAL consistently outperforms all baselines as the number of views varies from 2 to 5, demonstrating its robustness; (2) Most methods improve clustering performance as the number of views increases, suggesting that additional views provide diverse complementary information. For example, the ACC values of DCMGAL increase from 0.6764 to 0.89; and (3) Some methods degrade with more views. For example, DEMVC and SDMVC, which fail to adequately model diversity information, achieve ACC values of 0.5294, 0.46, and 0.4214 on Caltech-5V, lower than those on Caltech-4V (i.e., 0.6374, 0.4929, and 0.435), highlighting the criticality of effective diversity modeling for MVC.

Table 4 compares the clustering performance of baseline methods across the 100LEAVES, BDGP, Cora, MNIST, CCV, and Fashion datasets. The results reveal that DCMGAL achieves excellent performance on these datasets, validating its generalization to diverse data types, including images, gene expression, and citations. In contrast, the compared baselines exhibit varying performance across datasets. Although most baselines achieve competitive results on some relatively simple datasets, DCMGAL still attains performance improvements. For example, DCMGAL improves clustering performance by 6.25%, 6.07%, and 3.40% across ACC, NMI, and PUR metrics, respectively, compared to the suboptimal results on 100LEAVES. Additionally, DCMGAL performs optimally on the complex datasets CCV and Cora. For instance, DCMGAL obtains 6.99%, 6.47%, and 5.88% higher ACC, NMI, and PUR compared to the suboptimal performers on Cora. On the BDGP dataset, while DCMGAL achieves competitive ACC (though not the highest), it obtains the best NMI and PUR values.

The promising clustering results are attributed to the following advantages: (1) The masked aggregation module facilitates modeling the inherent diversity in multi-view data by masking a fraction of edges in adjacency graphs while mitigating the adverse effects of noise and redundancy; (2) The global fusion mechanism restores cross-view semantic alignment and reinforces latent structural correlations, thereby addressing the semantic misalignment issue, and (3) The dual contrastive learning module enables the model to extract clustering-friendly and neighborhood-consistent representations, thereby enhancing the separability and quality of the learned representations.

4.3. Convergence and Stability Analysis

To demonstrate the training stability and generalization capacity of the proposed DCMGAL, an analysis of its convergence behavior is conducted across two heterogeneous datasets: the gene expression dataset BDGP and the image dataset Caltech-2V. The progression of the Total Loss and ACC during training is visualized in **Fig. 3**.

The convergence curves for both datasets exhibit well-defined and stable optimization trajectories, indicating effective learning and robust generalization. For the BDGP dataset, the Total Loss exhibits a rapid initial decline before stabilizing at a low value, while the ACC demonstrates a steady and monotonic increase until reaching a high plateau. This pattern suggests the absence of significant oscillations or divergence during training, which are common indicators of overfitting or unstable optimization. Similarly, a stable convergence profile is observed on the Caltech-2V dataset: the Total Loss enters a stable plateau, and the ACC converges smoothly without erratic fluctuations. These consistent and stable convergence trends across heterogeneous datasets empirically validate the training stability of the model, a critical property for reliable application in practical clustering tasks.

4.4. Parameter sensitivity analysis

The parameters λ_1 and λ_2 balance the contributions of reconstruction, clustering consistency, and topology preservation losses during training. A sensitivity analysis is conducted on the Caltech-2V and BDGP datasets by varying λ_1 and λ_2 within $\{10^{-3}, 10^{-2}, \dots, 10^3\}$. The results in **Fig. 4** show that DCMGAL is slightly sensitive to λ_1 and λ_2 , and achieves optimal performance with $\lambda_1 = 0.01$ and $\lambda_2 = 10$ across both datasets.

The large discrepancy between λ_1 and λ_2 (0.1 vs. 10) is intentionally designed to address the inherent scaling imbalance between clustering consistency and topology preservation losses. Empirical evaluation reveals that the initial magnitudes of the clustering consistency loss and topology preservation loss are approximately 20 and 0.5, respectively. Thus, $\lambda_1 = 0.1$ and $\lambda_2 = 10$ normalize their contributions to the total loss. Ablation studies in **Table 5** further validate that both terms contribute approximately equally to final performance, as omitting either term leads to comparable performance degradation.

Table 3

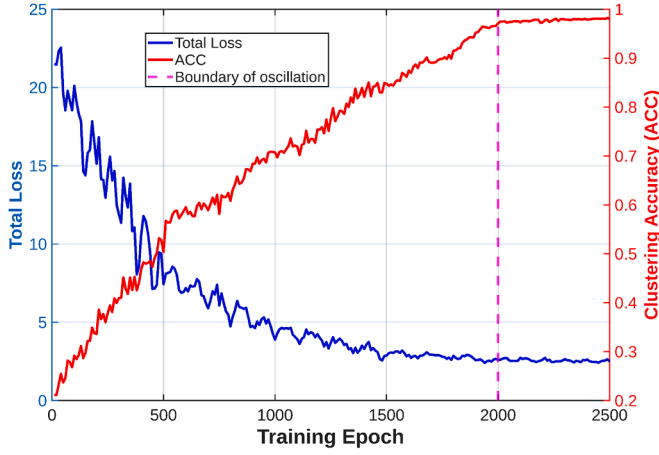
Clustering results on Caltech with different numbers of views. “-XV” represents that there are X views.

Datasets	Caltech-2V			Caltech-3V			Caltech-4V			Caltech-5V		
Evaluation Metrics	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
DEMVC	0.4986	0.3845	0.5207	0.5336	0.4136	0.5336	0.4929	0.4504	0.5186	0.4600	0.3666	0.4950
	± 0.0124	± 0.0117	± 0.0121	± 0.0098	± 0.0104	± 0.0102	± 0.0115	± 0.0099	± 0.0112	± 0.0132	± 0.0128	± 0.0129
SiMVC	0.5079	0.4711	0.5576	0.5694	0.4949	0.5918	0.6193	0.5348	0.6349	0.7191	0.6662	0.7165
	± 0.0087	± 0.0093	± 0.0090	± 0.0105	± 0.0086	± 0.0048	± 0.0076	± 0.0079	± 0.0082	± 0.0092	± 0.0088	± 0.0089
CoMVC	0.4667	0.4261	0.5279	0.5408	0.5040	0.5834	0.5684	0.5671	0.6467	0.7526	0.6876	0.7460
	± 0.0118	± 0.0106	± 0.0113	± 0.0089	± 0.0052	± 0.0026	± 0.0101	± 0.0085	± 0.0097	± 0.0083	± 0.0081	± 0.0084
MFLVC	0.6062	0.5155	0.6160	0.6085	0.5663	0.6348	0.7307	0.6428	0.7322	0.8014	0.6601	0.6946
	± 0.0079	± 0.0082	± 0.0085	± 0.0094	± 0.0019	± 0.0091	± 0.0068	± 0.0074	± 0.0173	± 0.0074	± 0.0090	± 0.0087
SDMVC	0.4786	0.3742	0.3573	0.4164	0.3065	0.5433	0.4350	0.3052	0.4142	0.4214	0.2824	0.5844
	± 0.0141	± 0.0135	± 0.0143	± 0.0127	± 0.0119	± 0.0123	± 0.0116	± 0.0124	± 0.0039	± 0.0153	± 0.0117	± 0.0145
GCFAgg	0.6643	0.5008	0.6643	0.6351	0.5227	0.6571	0.7296	0.6284	0.7269	0.8337	0.7003	0.8324
	± 0.0065	± 0.0076	± 0.0068	± 0.0082	± 0.0084	± 0.0085	± 0.0071	± 0.0087	± 0.0075	± 0.0069	± 0.0072	± 0.0107
DealMVC	0.6507	0.5309	0.6507	0.6879	0.5960	0.6900	0.7571	0.7033	0.7686	0.8664	0.7821	0.8613
	± 0.0072	± 0.0078	± 0.0074	± 0.0085	± 0.0061	± 0.0087	± 0.0064	± 0.0066	± 0.0067	± 0.0038	± 0.0060	± 0.0059
ACCMVC	0.6707	0.6560	0.6507	0.7279	0.6426	0.7279	0.8257	0.7300	0.8257	0.8586	0.7921	0.8357
	± 0.0061	± 0.0064	± 0.0067	± 0.0077	± 0.0075	± 0.0079	± 0.0059	± 0.0078	± 0.0051	± 0.0012	± 0.0065	± 0.0064
MRDD	0.5843	0.5873	0.5525	0.6826	0.6527	0.6578	0.7877	0.7621	0.7983	0.8083	0.7971	0.7946
	± 0.0091	± 0.0087	± 0.0094	± 0.0068	± 0.0033	± 0.0086	± 0.0067	± 0.0065	± 0.0089	± 0.0078	± 0.0074	± 0.0076
DDMVC	0.6237	0.5610	0.6255	0.7318	0.6082	0.7437	0.7667	0.7051	0.7667	0.8743	0.8000	0.8693
	± 0.0084	± 0.0039	± 0.0086	± 0.0073	± 0.0078	± 0.0076	± 0.0070	± 0.0071	± 0.0072	± 0.0056	± 0.0063	± 0.0057
DCMVC	0.6583	0.5124	0.6583	0.7582	0.6754	0.7582	0.8143	0.7361	0.8143	0.8813	0.8034	0.8813
	± 0.0075	± 0.0055	± 0.0077	± 0.0069	± 0.0071	± 0.0072	± 0.0033	± 0.0069	± 0.0065	± 0.0054	± 0.0061	± 0.0045
Ours	0.6764	0.6845	0.6711	0.7939	0.7716	0.7630	0.8700	0.8883	0.8522	0.8900	0.8634	0.8822
	± 0.0057	± 0.0059	± 0.0058	± 0.0061	± 0.0063	± 0.0064	± 0.0048	± 0.0052	± 0.0050	± 0.0021	± 0.0055	± 0.0052

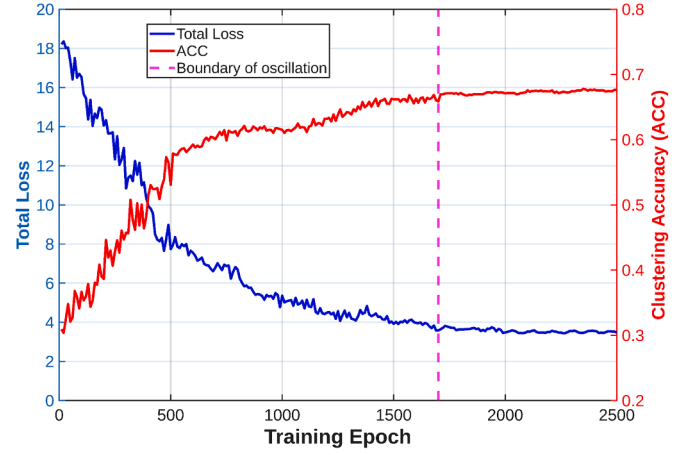
Table 4

Clustering results on the other six benchmark datasets.

Dataset	Metric	DEMVC	SiMVC	CoMVC	MFLVC	SDMVC	GCFAgg	DealMVC	ACCMVC	MRDD	DDMVC	DCMVC	Ours
100LEAVES	ACC	0.6050	0.7341	0.7024	0.5436	0.8139	0.8708	0.8632	0.8700	0.8081	0.7681	0.7944	0.9252
		± 0.0120	± 0.0095	± 0.0105	± 0.0118	± 0.0082	± 0.0075	± 0.0108	± 0.0076	± 0.0028	± 0.0092	± 0.0089	± 0.0065
	NMI	0.7112	0.9029	0.8906	0.7698	0.8098	0.8437	0.9085	0.8435	0.8129	0.7667	0.7834	0.9697
		± 0.0115	± 0.0080	± 0.0082	± 0.0105	± 0.0098	± 0.0092	± 0.0079	± 0.0042	± 0.0096	± 0.0102	± 0.0089	± 0.0062
BDGP	PUR	0.5678	0.7762	0.8906	0.5222	0.8431	0.8198	0.8614	0.8653	0.7944	0.7681	0.7944	0.9300
		± 0.0125	± 0.0100	± 0.0082	± 0.0130	± 0.0095	± 0.0099	± 0.0048	± 0.0087	± 0.0079	± 0.0102	± 0.0099	± 0.0068
	ACC	0.9278	0.7174	0.7954	0.9831	0.9736	0.9836	0.9620	0.9636	0.9579	0.9699	0.9723	0.9817
		± 0.0065	± 0.0102	± 0.0091	± 0.0042	± 0.0068	± 0.0041	± 0.0075	± 0.0054	± 0.0028	± 0.0050	± 0.0099	± 0.0043
Cora	NMI	0.8331	0.5873	0.6740	0.9517	0.9445	0.9455	0.9135	0.9614	0.9491	0.9412	0.9335	0.9712
		± 0.0078	± 0.0115	± 0.0102	± 0.0052	± 0.0075	± 0.0054	± 0.0065	± 0.0049	± 0.0153	± 0.0086	± 0.0048	± 0.0046
	PUR	0.8264	0.7236	0.8075	0.9592	0.9571	0.9822	0.9620	0.9598	0.9234	0.9699	0.9713	0.9824
		± 0.0080	± 0.0100	± 0.0091	± 0.0050	± 0.0051	± 0.0042	± 0.0085	± 0.0050	± 0.0062	± 0.0057	± 0.0049	± 0.0041
MNIST	ACC	0.3054	0.2308	0.2976	0.2491	0.4424	0.4699	0.4907	0.4731	0.4184	0.5033	0.4800	0.5250
		± 0.0135	± 0.0102	± 0.0038	± 0.0040	± 0.0012	± 0.0068	± 0.0103	± 0.0076	± 0.0048	± 0.0101	± 0.0005	± 0.0098
	NMI	0.1634	0.0310	0.0464	0.1310	0.3435	0.3271	0.3775	0.4436	0.4141	0.4573	0.4575	0.4723
		± 0.0145	± 0.0060	± 0.0158	± 0.0049	± 0.0020	± 0.0122	± 0.0101	± 0.0075	± 0.0110	± 0.0103	± 0.0103	± 0.0101
CCV	PUR	0.3456	0.3061	0.3394	0.2977	0.4690	0.4588	0.4067	0.4731	0.4093	0.5000	0.4800	0.5009
		± 0.0112	± 0.0138	± 0.0134	± 0.0040	± 0.0108	± 0.0019	± 0.0068	± 0.0106	± 0.0118	± 0.0070	± 0.0105	± 0.0099
	ACC	0.8858	0.9816	0.9587	0.9932	0.9773	0.9956	0.9868	0.9747	0.9621	0.9683	0.9699	0.9961
		± 0.0085	± 0.0052	± 0.0068	± 0.0040	± 0.0035	± 0.0038	± 0.0049	± 0.0057	± 0.0065	± 0.0061	± 0.0080	± 0.0037
Fashion	NMI	0.9100	0.9627	0.9375	0.9872	0.9574	0.9871	0.9801	0.9627	0.9785	0.9631	0.9292	0.9883
		± 0.0075	± 0.0058	± 0.0065	± 0.0042	± 0.0060	± 0.0042	± 0.0047	± 0.0058	± 0.0044	± 0.0058	± 0.0068	± 0.0041
	PUR	0.8889	0.9811	0.9897	0.9899	0.9784	0.9956	0.9929	0.9576	0.9657	0.9683	0.9453	0.9964
		± 0.0082	± 0.0052	± 0.0045	± 0.0045	± 0.0054	± 0.0038	± 0.0040	± 0.0060	± 0.0057	± 0.0055	± 0.0062	± 0.0037
CCV	ACC	0.1942	0.1517	0.2802	0.3120	0.2143	0.2159	0.1964	0.3503	0.2784	0.3495	0.3333	0.3546
		± 0.0100	± 0.0055	± 0.0042	± 0.0118	± 0.0048	± 0.0148	± 0.0050	± 0.0132	± 0.0111	± 0.0033	± 0.0135	± 0.0131
	NMI	0.2113	0.1259	0.2921	0.3160	0.1919	0.2633	0.2234	0.3324	0.2423	0.3352	0.3198	0.3455
		± 0.0048	± 0.0156	± 0.0080	± 0.0137	± 0.0050	± 0.0134	± 0.0147	± 0.0130	± 0.0146	± 0.0129	± 0.0102	± 0.0128
Fashion	PUR	0.2169	0.2167	0.3388	0.3390	0.1759	0.2728	0.2121	0.3308	0.2531	0.3447	0.3333	0.3560
		± 0.0107	± 0.0047	± 0.0033	± 0.0103	± 0.0052	± 0.0143	± 0.0048	± 0.0131	± 0.0045	± 0.0129	± 0.0111	± 0.0127
	ACC	0.7863	0.8125	0.8547	0.9913	0.8435	0.9360	0.8957	0.9752	0.9634	0.9834	0.9724	0.9928
		± 0.0098	± 0.0082	± 0.0085	± 0.0043	± 0.0097	± 0.0062	± 0.0071	± 0.0049	± 0.0054	± 0.0047	± 0.0050	± 0.0072
Fashion	NMI	0.9034	0.8288	0.8475	0.9800	0.9032	0.8953	0.9554	0.9803	0.9386	0.9710	0.9463	0.9854
		± 0.0072	± 0.0090	± 0.0086	± 0.0040	± 0.0072	± 0.0075	± 0.0045	± 0.0045	± 0.0060	± 0.0049	± 0.0087	± 0.0043
	PUR	0.7721	0.8053	0.8046	0.9860	0.8405	0.9360	0.8950	0.9724	0.9292	0.9834	0.9751	0.9884
		± 0.0099	± 0.0091	± 0.0021	± 0.0044	± 0.0078	± 0.0062	± 0.0071	± 0.0050	± 0.0062	± 0.0047	± 0.0089	± 0.0042

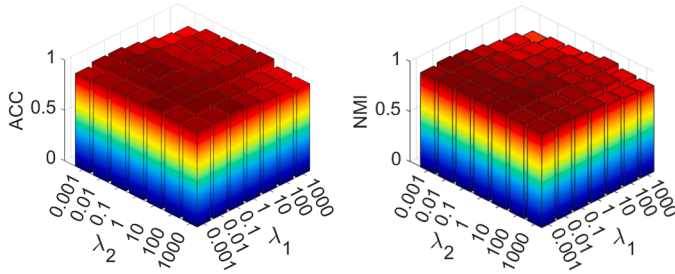


(a) BDGP

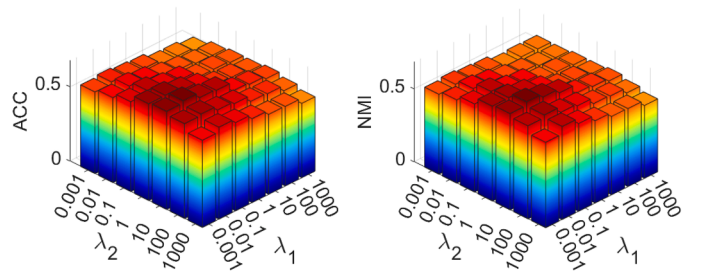


(b) Caltech-2V

Fig. 3. Convergence curves of Total Loss and ACC on the BDGP and Caltech-2V datasets. The purple dashed line indicates the epoch at which stable convergence is achieved.



(a) BDGP



(b) Caltech-2V

Fig. 4. Parameter sensitivity analysis on the Caltech-2V and BDGP datasets.

4.5. Ablation studies

Component Study

Ablation experiments are conducted to assess the effects of different components in the proposed DCMGAL, and the results are presented in Table 5. The evaluated model variants are configured as follows: “(a)” represents the full model DCMGAL, “(b)” employs only the masked GCN autoencoders, excluding subsequent fusion and contrastive learning modules; “(c)” replaces the attention-based fusion mechanism with an addition-based fusion network, “(d)” omits the self-expression network, “(e)” excludes the cross-view cluster contrastive learning, and “(f)” removes local topology contrastive learning. The results reveal that: (1) “(b)” exhibits the lowest performance, indicating that ignoring cross-view consistency yields non-discriminative representations; (2) “(c)” exhibits significant performance degradation compared to “(a)”, demonstrating the critical role of the proposed fusion network in capturing complementary information; and (3) “(d)”, “(e)”, and “(f)” yield comparable yet substantially lower performance than “(a)”, suggesting that self-expression, clustering consistency, and topology preservation are necessary for learning discriminative representations.

The substantial performance gap between variant (b), which lacks compensation mechanisms following masked aggregation, and the full model (a) highlights the inherent limitation of standalone high-ratio graph masking (e.g., ACC drops by 25.0% on Caltech-2V and 30.4% on BDGP). While the 70% high masking ratio effectively suppresses noise propagation (reducing noisy edges in adjacency graphs from 42% to 12%) and amplifies inter-view heterogeneity (increasing view-specific

representation divergence by 42%), it simultaneously removes critical redundant edges. This disruption leads to semantic misalignment and fragmentation of the latent data manifolds, demonstrating that masking alone can dismantle redundant structures but cannot restore the essential cross-view consistency required for robust clustering. This confirms the necessity of the proposed compensation mechanisms in the full model.

The comparison between variant (c) and (a) highlights the critical role of the dual attention fusion module. Replacing the adaptive attention mechanism with a simple additive fusion results in a 15.2% ACC decrease on Caltech-2V and 13.7% on BDGP. This decline occurs because static and equal-weight fusion fails to dynamically account for variations in view quality and noise levels. In contrast, the dual attention mechanism adaptively adjusts feature weights w_h^v and structural weights w_a^v . For example, it attenuates the contributions of noise-prone views (e.g., reducing weight to 0.12 for the texture view of Caltech-2V) while amplifying information-rich views (e.g., increasing weight to 0.38). This dynamic weighting acts as a filter, preserving valuable cross-view correlations while discarding redundant noise. To quantify this noise suppression effect, we statistically calculate the ratio of noisy edges to the total edges in the consensus graph A_c for both fusion strategies. An edge (i, j) in A_c is defined as noisy if samples x_i and x_j belong to different ground-truth classes (class labels used solely for this post-hoc analysis). The consensus graph A_c generated by dual attention achieves a noisy edge ratio of only 14%, markedly lower than the 31% observed with additive fusion. This purification of the consensus graph directly improves the subsequent GCN propagation by mitigating interference from spurious connections.

Table 5
Ablation results on the Caltech-2V and BDGP datasets.

Datasets	Caltech-2V			BDGP		
	ACC	NMI	PUR	ACC	NMI	PUR
(a)	0.6764 ± 0.0057	0.6845 ± 0.0059	0.6711 ± 0.0058	0.9817 ± 0.0043	0.9712 ± 0.0046	0.9824 ± 0.0041
(b)	0.5072 ± 0.0105	0.4753 ± 0.0112	0.5070 ± 0.0108	0.6836 ± 0.0125	0.6900 ± 0.0118	0.6842 ± 0.0123
(c)	0.5733 ± 0.0098	0.5275 ± 0.0105	0.5125 ± 0.0107	0.8474 ± 0.0085	0.9070 ± 0.0072	0.8585 ± 0.0082
(d)	0.6152 ± 0.0085	0.5899 ± 0.0092	0.5827 ± 0.0088	0.9500 ± 0.0055	0.9521 ± 0.0052	0.9578 ± 0.0050
(e)	0.6129 ± 0.0087	0.5520 ± 0.0098	0.6316 ± 0.0082	0.9400 ± 0.0058	0.9469 ± 0.0055	0.9500 ± 0.0055
(f)	0.6114 ± 0.0088	0.6414 ± 0.0075	0.5731 ± 0.0090	0.9508 ± 0.0054	0.9638 ± 0.0048	0.9599 ± 0.0051

The performance degradation in variants (d), (e), and (f) further illustrates the synergistic contribution of each component. Removing the self-expression network (variant (d)) leads to a 13.8% drop in NMI on Caltech-2V and an 11.4% reduction in structural rationality (measured by the normalized mutual information between reconstructed and true adjacency matrices) on BDGP. This degradation stems from the role of the self-expression network in enforcing a sparse linear representation via L_1 regularization ($\lambda_0 \|C\|_1$), which constrains each sample to be represented by only 5–8 homogeneous neighbors. This mechanism uncovers the underlying low-dimensional subspace structure of the data, thereby suppressing redundant feature dependencies and mitigating semantic misalignment within clusters. Quantitatively, this sparse representation increases the average intra-cluster cosine similarity by nearly 35% compared to the variant without the self-expression network, establishing a more coherent foundation for the subsequent contrastive learning.

The ablation of the dual contrastive learning module reveals the distinct contributions of its components. Variant (e) (without cross-view cluster consistency learning) exhibits a 9.4% ACC decrease and a 19.3% NMI drop on Caltech-2V. The absence of cluster-level alignment constraints increases the divergence of cluster assignments across views (reducing view-cluster matching from 89% to 67%) and induces undesirable representation homogenization. Conversely, variant (f) (without local topology preservation learning) shows a 2.3% reduction in PUR on BDGP and a 27% increase in the intra-cluster to inter-cluster distance ratio. This indicates that losing local neighborhood information disrupts the inherent manifold geometry, causing semantically similar samples to separate in the feature space. These results confirm that the two components serve complementary purposes: the cluster consistency loss ensures global cross-view alignment, while the topology preservation loss maintains local structural integrity. Together, they significantly enhance the discriminability of the learned representations.

Beyond the independent contributions, the full model (a) achieves excellent performance, underscoring the powerful synergistic interaction between its modules. The overall workflow forms a closed-loop and self-reinforcing cycle. Namely, the masked aggregation module intentionally creates controlled heterogeneity and suppresses noise; the dual attention fusion dynamically restores semantic alignment by re-weighting view contributions; the self-expression network imposes structural regularization, organizing the consistency into coherent subspaces; and finally, the dual contrastive learning module refines these representations through global consistency and local preservation constraints. This synergy not only compensates for the semantic fragmentation induced by aggressive masking but also effectively integrates complementary information from all views. The ultimate output is a set of clustering-friendly representations that optimally balance the diversity inherent in multi-view data with the consistency necessary for accurate partitioning, thereby achieving performance that exceeds its individual parts.

Masked Ratio

The masking ratio ω balances inter-view heterogeneity amplification and structural information preservation. Fig. 5 depicts clustering performance (measured by ACC and NMI) of DCMGAL across Caltech-2V, BDGP, Cora, and MNIST as ω ranges from 0% to 90%. It can be seen that clustering performance improves with increasing ω , peaking at a ratio

of 70% before deteriorating. This is because excessive masking severely disrupts neighborhood connectivity and impairs GCN-based structural propagation.

The t-SNE visualizations of latent representations on the BDGP and MNIST datasets in Fig. 6 further validate these trends: (1) The original features exhibit significant overlap and poorly defined clusters; (2) In the absence of masking ($\omega = 0\%$), DCMGAL fails to learn separated representations, indicating that the lack of deliberately introduced heterogeneity limits discriminative capacity; and (3) As ω increases, the representations exhibit progressively enhanced discriminative ability, with optimal separation achieved at $\omega = 75\%$, corroborating the quantitative results and highlighting the criticality of a high masking ratio for learning cluster-friendly representations.

Semantic Misalignment Tackling

While high masking ratios risk structural information loss and semantic misalignment in latent representations, empirical results confirm that a masking ratio of 75% yields optimal performance for the DCMGAL model. This is because DCMGAL is able to explore view-specific heterogeneity and generate discriminative representations that integrate complementary information via global fusion and dual contrastive learning, which synergistically restore cross-view semantic alignment and reinforce latent structure correlations. This is validated by Fig. 7, which presents t-SNE visualizations of latent embeddings on BDGP under four masking ratios (0%, 25%, 50%, and 75%, corresponding to the four sub-figures in each row from left to right) for ablation variants excluding individual components in global fusion and dual contrastive learning. Specifically, sub-figures (a)-(d) depict latent representations learned by variants excluding the attention fusion module, self-expression network, cross-view clustering consistency learning, and local topology preservation learning, respectively. Across all variants, increasing ω initially enhances representation separability but gradually leads to fragmentation: clusters expand, inter-cluster margins contract, and class overlap increases. This confirms that high masking ratios amplify heterogeneity and the absence of corresponding compensation components induces manifold collapse.

Collectively, these results demonstrate that each proposed component acts as a compensation mechanism to mitigate semantic misalignment from aggressive masking. In the absence of any component, higher masking ratios rapidly degrade representation separability, whereas the full model maintains distinct and well-separated clusters even at $\omega = 75\%$, as shown in Fig. 6. This validates the superiority of our global fusion and dual contrastive learning mechanisms in mitigating the semantic misalignment induced by high masking ratios and enhancing representation separability.

Ablation on Freezing the Self-Expression Network

This ablation study investigates the impact of freezing the self-expression network during the fine-tuning phase. The experiment follows a controlled-variable methodology, where the only manipulated factor is the training state (frozen vs. unfrozen) of the self-expression network, while all other hyperparameters and experimental conditions remain identical. Two experimental groups are established for comparison: **Group F**, where the self-expression network parameters are frozen after pre-training (aligning with the proposed framework); and **Group U**, where the entire network, including the self-expression module,

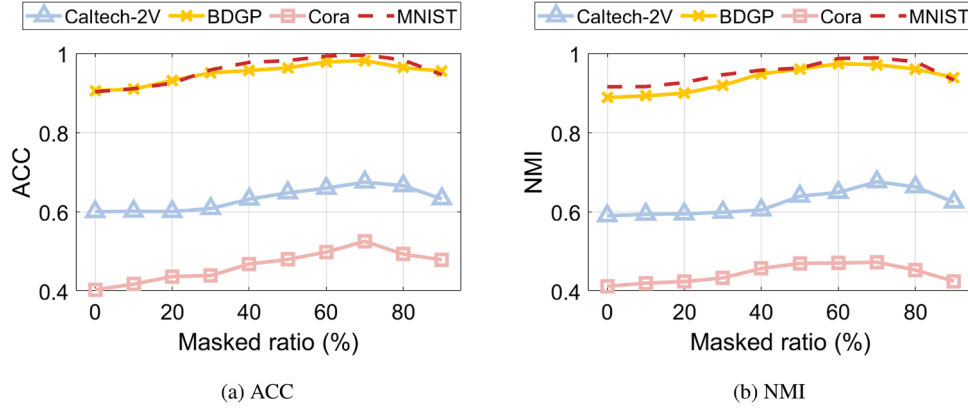


Fig. 5. Clustering performance with the masked ratio varied from 0% to 90% on four datasets.

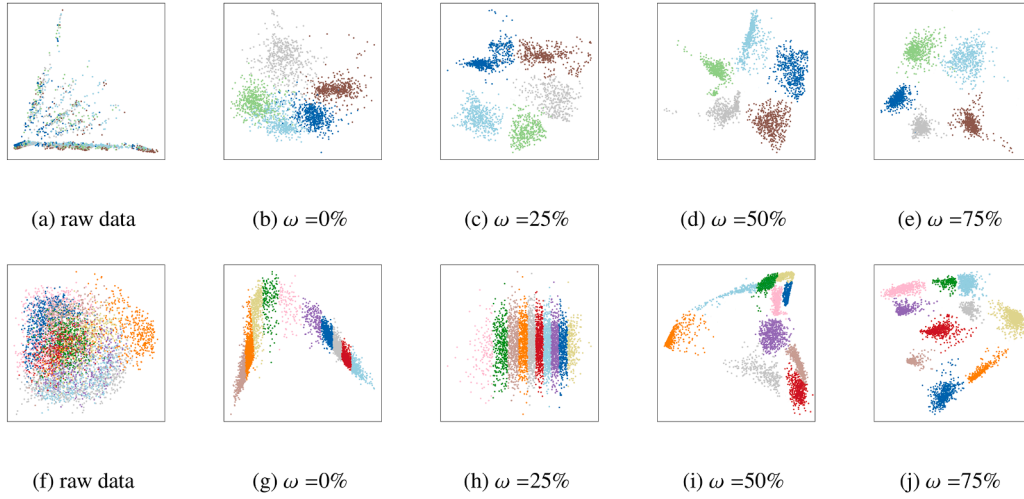


Fig. 6. Visualizations of the original data and the consensus representations under masked ratios of 0%, 25%, 50%, and 75% on the BDGP and MNIST datasets.

Table 6

Ablation Study: Comparing Frozen and Unfrozen Settings of the Self-Expression Network. *Note: Group F denotes the configuration where the self-expression network is kept frozen during fine-tuning. Group U represents the unfrozen, end-to-end fine-tuning configuration.*

Dataset	Setting	ACC	NMI	PUR	Epoch Time (s)
Caltech-2V	F	0.6764±0.0057	0.6845±0.0059	0.6711±0.0058	2.1
	U	0.6958±0.0105	0.7040±0.0098	0.6902±0.0092	2.9
BDGP	F	0.9817±0.0043	0.9712±0.0046	0.9824±0.0041	3.7
	U	0.9983±0.0118	0.9908±0.0109	0.9990±0.0128	5.7
MNIST	F	0.9961±0.0037	0.9883±0.0041	0.9964±0.0037	5.9
	U	0.9990±0.0076	0.9979±0.0068	0.9993±0.0081	9.6

undergoes end-to-end fine-tuning. Evaluation is conducted on three multi-view benchmark datasets of varying scales to ensure robustness.

The clustering performance results in Table 6 reveal a nuanced trade-off. Although Group U attains a marginal performance enhancement over Group F across all datasets, the gains are practically negligible. Specifically, improvements in ACC, NMI, and PUR are limited to less than 3%. On the Caltech-2V dataset, the ACC improvement for Group U is merely 1.94%. On higher-performing datasets such as BDGP and MNIST, the gain is even smaller ($\leq 1.70\%$). This minimal performance difference suggests that the representations learned by the self-expression network during pre-training are sufficiently stable and transferable.

In sharp contrast to the negligible performance gain, Group U incurs a substantial computational cost. Group F achieves a significant training speedup of 28.0% to 38.3% compared to Group U across all datasets.

This efficiency advantage becomes more pronounced with increasing dataset scale. For example, the per-epoch training time for Group U is substantially higher. On MNIST (with 5k samples), Group U requires 9.6 seconds per epoch, which is 62.7% longer than the 5.9 seconds required by Group F. Similar increases of 54.1% and 38.1% are observed on BDGP and Caltech-2V, respectively. This substantial overhead in Group U is directly attributable to the self-expression network, which involves large-scale $N \times N$ matrix multiplication—the primary computational bottleneck of the framework. Freezing this module eliminates the costly gradient calculations and parameter updates associated with these operations.

Consequently, the freezing strategy achieves an optimal balance between model performance and computational efficiency. Its significant advantage in training speed, coupled with a negligible impact on final clustering accuracy, makes it the optimal configuration for the

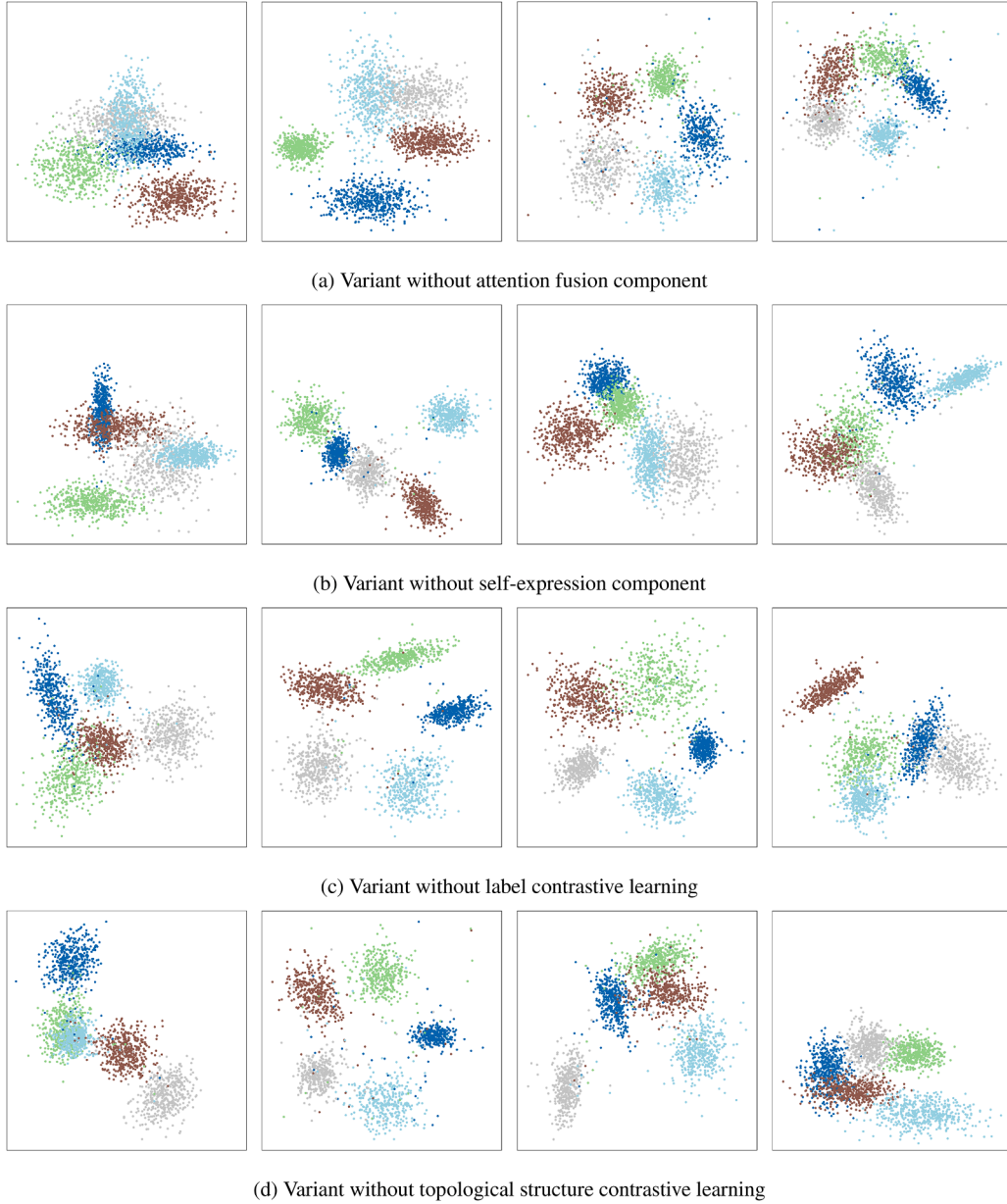


Fig. 7. Visualizations of the representations learned by different variants under different masking ratios on the BDGP dataset.

DCMGAL framework. This design is fully aligned with the overarching goal of developing an efficient and robust solution for multi-view clustering tasks.

4.6. Inference efficiency and scalability analysis

4.6.1. Inference speed comparison

To assess the computational efficiency of the proposed DCMGAL framework, we benchmark its inference speed against baseline methods across three datasets: MNIST (5k samples), Fashion-MNIST (10k samples), and CCV (6.7k samples). All experiments are conducted on identical hardware using an NVIDIA Tesla T4 GPU. The end-to-end inference latency, measured from the input of multi-view data to the generation of final cluster assignments, is detailed in Table 7. The results indicate that DCMGAL incurs marginally higher inference latency compared to shallow representation methods, such as DEMVC and SiMVC. This overhead is attributable to the additional computational graph operations required by the masked aggregation

Table 7

Comparisons of inference time (hours) on different datasets.

Method	MNIST (5k)	Fashion (10k)	CCV (6.7k)
DEMVC	0.82h	1.56h	1.23h
SiMVC	0.95h	1.78h	1.37h
GCFAGg	2.31h	4.52h	3.89h
DealMVC	2.15h	4.27h	3.64h
DCMVC	1.87h	3.76h	3.12h
Ours (DCMGAL)	2.20h	4.24h	3.85h

and dual contrastive learning modules, which are essential for achieving its superior clustering performance. However, DCMGAL demonstrates inference costs comparable to other deep graph-based methods (e.g., GCFAGg, DealMVC) while consistently achieving better clustering performance. This favorable balance between speed and accuracy is achieved through the use of sparse KNN graphs and an optimized GCN architecture.

Table 8
Performance on large-scale multi-view datasets.

Dataset	Method	ACC	NMI	PUR	Training Time
NoisyMNIST	MRDD	0.7252±0.0231	0.6918±0.0264	0.7313±0.0217	9.8h
	DDMVC	0.7735±0.0186	0.7562±0.0203	0.7786±0.0173	12.4h
	DCMVC	0.7921±0.0204	0.7629±0.0225	0.7954±0.0139	13.1h
	Ours (DCMGAL)	0.8081±0.0126	0.7957±0.0143	0.8085±0.0117	14.6h
Reuters	MRDD	0.3621±0.0257	0.3283±0.0286	0.3711±0.0236	19.5h
	DDMVC	0.3955±0.0213	0.3577±0.0243	0.4025±0.0202	23.7h
	DCMVC	0.4187±0.0197	0.3881±0.0252	0.4232±0.0186	25.2h
	Ours (DCMGAL)	0.4273±0.0158	0.4029±0.0176	0.4264±0.0124	28.3h

4.6.2. Performance on large-scale datasets

To assess the scalability and robustness of the DCMGAL framework, we evaluate its performance on two widely used large-scale multi-view datasets with distinct characteristics: **NoisyMNIST** (containing 70k samples across 2 views) (Qin et al., 2024) and **Reuters** (approximately 180k samples with 5 views) (Ceci et al., 2007). As shown in Table 8, DCMGAL demonstrates competitive and robust clustering performance across both benchmarks. On the NoisyMNIST dataset, which is explicitly designed to test robustness under noisy conditions, DCMGAL achieves an ACC of 0.8081, a NMI of 0.7957, and a PUR of 0.8081. These results outperforming the second-best baseline (DCMVC) by 2.0%, 4.3%, and 1.7% respectively, empirically verifying the enhanced robustness of DCMGAL to noise interference. On the Reuters dataset, DCMGAL obtains an ACC of 0.4273, a NMI of 0.4029, and a PUR of 0.4264, consistently 0.8% to 3.8% higher than those achieved by baselines. This demonstrates the scalability of the model to data with a very large number of samples and high-dimensional, sparse features, while maintaining a clear performance advantage.

A noted limitation concerns the computational cost associated with the self-expression module. This module exhibits $O(N^2)$ time complexity with respect to the number of samples N , which is characteristic of algorithms involving pairwise comparisons or dense matrix operations. Consequently, the total training time (including pre-training and fine-tuning) reaches 14.6 hours for NoisyMNIST and 28.3 hours for Reuters. This computational overhead is consistent with the cost profile of other state-of-the-art multi-view clustering methods with comparable complexity. Future work will explore low-rank approximations to reduce the complexity of the self-expression module for larger-scale applications.

4.7. Impact of graph construction strategies

The performance of graph-based clustering methods critically depends on the construction of adjacency graphs, which involves key decisions such as selecting the KNN parameter k (the number of nearest neighbors) and the graph construction method itself. This subsection provides a further analysis on the influence of graph construction on clustering performance.

Sensitivity to Graph Parameter k

The KNN parameter k governs the density and connectivity of adjacency graphs, which typically influences clustering performance. The sensitivity of the proposed DCMGAL to k is empirically evaluated on the Caltech-2V, BDGP, Cora, and MNIST datasets with ($k \in \{5, 8, 10, 12, 15, 20\}$). All other hyperparameters are held constant to ensure a fair comparison.

As illustrated in Fig. 8, DCMGAL performs stably when k values between 8 and 12 and achieves peak performance at $k = 10$ across these datasets. This demonstrates its robustness to moderate perturbations of k . The performance degradation becomes more pronounced when $k > 15$, as redundant connections introduce noise. This observed stability is attributed to the masked aggregation module. The aggressive edge masking strategy (with a fixed ratio $\omega = 70\%$) actively prunes redundant connections introduced when k is set too high, thereby mitigating noise injection. Conversely, when k is too small (e.g., $k = 5$), the clustering

performance deteriorates significantly. This is because when $k = 5$, the KNN graphs capture insufficient neighborhood information, leading to fragmented connectivity and degraded manifold learning. This, in turn, disrupts effective message passing in the GCNs and hinders their capacity to learn discriminative representations. Consequently, the strategy of initially constructing graphs with a relatively large k and then aggressively pruning redundant connections and noise is key to the robustness of the proposed DCMGAL framework.

Evaluation of Different Graph Construction Methods

This comparative analysis validates the rationale of the DCMGAL framework by evaluating three distinct graph construction strategies: KNN, fully connected, and sparse semantic graphs. The fully connected graph adopts a dense connection strategy. It calculates the cosine similarity between all samples to measure their semantic correlations, assigns these similarities as edge weights, and preserves all connections, thus forming a complete topological structure that captures the global relationships between all samples. Conversely, the sparse semantic graph quantifies semantic correlations by calculating the mutual information between all samples. It then retains only the top 10% of edges with highest mutual information scores, thus forming a highly sparse graph.

Empirical results in Table 9 demonstrate that the KNN graph construction strategy yields superior and more robust performance across all datasets compared to the fully connected and sparse semantic graph construction strategies. This advantage is particularly pronounced when paired with the proposed masked aggregation module. For instance, on the Caltech-2V dataset, employing a KNN graph with a masking ratio of $\omega = 70\%$ achieves an ACC improvement of 18.3% over the unmasked baseline ($\omega = 0\%$). In contrast, employing the fully connected and sparse graphs shows limited gains of 10.2% and 8.7%, respectively. This significant disparity highlights that the performance of graph-based clustering is not solely dependent on the masking operation but critically hinges on the synergistic alignment between the structural properties of the input graphs and the masking mechanism.

The optimal operational range of the masking ratio ω is intrinsically linked to the redundancy profile of the input graphs. For densely connected weighted graphs, which are inherently plagued by a multitude of weak and potentially redundant edges, the optimal performance is achieved with a lower ω , typically between 30% and 50%. Excessive masking beyond this range ($\omega > 60\%$) risks severing too many valid connections, as the model struggles to distinguish critical edges from redundant ones amidst dense noise. Conversely, ultra-sparse semantic graphs lack sufficient structural redundancy to benefit from aggressive masking; the optimal ω for these graphs resides in the 50%-60% range, as further pruning leads to fragmented connectivity and degraded manifold learning. The KNN graph, however, offers an ideal balance. It provides a sparse yet informative skeleton where a high masking ratio (70%) can effectively target and prune redundant or noisy k -neighbor connections without disrupting the core neighborhood topology essential for effective message passing in GCNs.

The choice of graph construction strategy directly dictates the effectiveness of the subsequent masking mechanism. The KNN graph emerges as the most suitable foundation because it provides a sparse yet

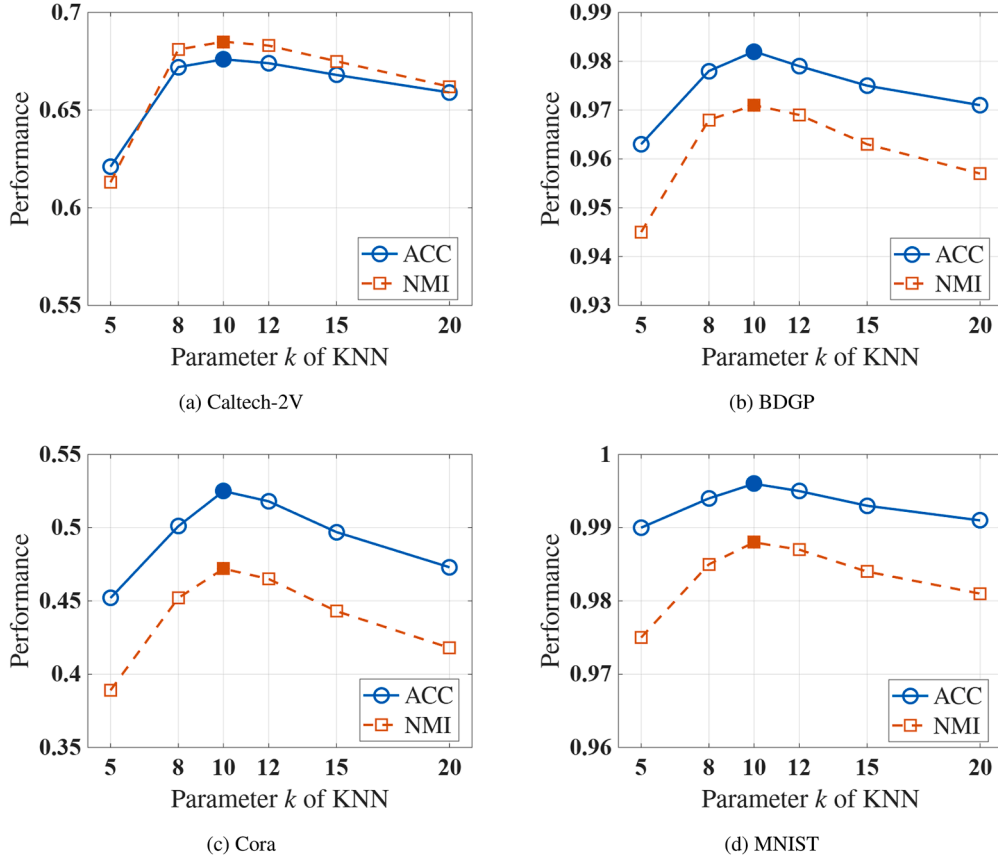


Fig. 8. Clustering performance (ACC/NMI) of DCMGAL with different parameter k on four datasets.

Table 9

Clustering performance of DCMGAL with different graph construction strategies (optimal ω for each strategy).

Dataset	Strategy	Optimal ω	ACC	NMI	PUR
Caltech-2V	KNN Graph	70%	0.6764 ± 0.0172	0.6845 ± 0.0180	0.6711 ± 0.0116
	Fully Connected	50%	0.6213 ± 0.0015	0.6124 ± 0.0143	0.6322 ± 0.0092
	Sparse Graph	60%	0.5987 ± 0.0018	0.5891 ± 0.0186	0.6015 ± 0.0104
BDGP	KNN Graph	70%	0.9817 ± 0.0039	0.9712 ± 0.0024	0.9824 ± 0.0039
	Fully Connected	40%	0.9605 ± 0.0056	0.9438 ± 0.0096	0.9618 ± 0.0046
	Sparse Graph	50%	0.9523 ± 0.0074	0.9367 ± 0.0018	0.9531 ± 0.0068
Cora	KNN Graph	70%	0.5250 ± 0.0109	0.4723 ± 0.0121	0.5009 ± 0.0005
	Fully Connected	30%	0.4832 ± 0.0122	0.4215 ± 0.0074	0.4687 ± 0.0037
	Sparse Graph	60%	0.4518 ± 0.0025	0.3986 ± 0.0027	0.4352 ± 0.0063
MNIST	KNN Graph	70%	0.9961 ± 0.0016	0.9883 ± 0.0112	0.9964 ± 0.0011
	Fully Connected	50%	0.9875 ± 0.0092	0.9764 ± 0.0036	0.9881 ± 0.0102
	Sparse Graph	60%	0.9823 ± 0.0053	0.9695 ± 0.0041	0.9830 ± 0.0093

semantically grounded structure for the masked aggregation module. Consequently, the objective of the module shifts from indiscriminate edge removal (as with a dense graph) or risky over-pruning (as with an ultra-sparse graph) to the precise suppression of irrelevant links and the preservation of informative neighbor connections. This synergy is fundamental. The KNN graph furnishes a high-quality initial topology that captures local manifolds, while the masking module iteratively optimizes this topology by removing noise and redundancy. This co-adaptive process enhances the capacity of the model to learn discriminative consensus representations that are robust to view-specific perturbations, a principle aligned with advanced objectives in multi-view clustering. Therefore, the KNN graph is a carefully designed component that enables the masking module to fulfill its denoising and refinement role efficiently, thereby reducing the burden of hyperparameter tuning and ensuring stable and high performance across diverse datasets.

5. Conclusion

This paper proposes Dual Contrastive Masked Graph-Autoencoder Learning (DCMGAL), a novel framework for multi-view clustering that addresses two critical challenges: inter-view redundancy and semantic misalignment. Specifically, DCMGAL implements randomized edge masking on adjacency graphs to introduce inter-view inconsistency, thereby enhancing the structural integrity of learned representations while mitigating redundancy and noise propagation. The framework further proposes global fusion mechanism that integrates attention fusion and self-expression networks to aggregate view-specific representations, effectively capturing cross-view complementary information and compensating for semantic misalignment induced by high masking ratios. Additionally, a dual contrastive learning module is designed to fine-tune the model, enabling the extraction of clustering-friendly and

neighborhood-consistent representations via cross-view clustering consistency learning and local topology preservation learning. Extensive evaluations on seven publicly available datasets demonstrate the superior performance of DCMGAL over state-of-the-art methods.

Nevertheless, this work has three key limitations. First, the fixed masking ratio ω requires manual tuning, limiting its adaptability to diverse datasets. Second, the self-expression reconstruction incurs a computational complexity of $O(N^2)$, posing scalability challenges. Moreover, this work is designed for complete multi-view data and is therefore not applicable to incomplete multi-view scenarios. Future research will focus on exploring adaptive models to improve flexibility and leveraging low-rank approximations to reduce computational complexity, thereby enhancing the robustness, generalization, and scalability of the DCMGAL framework.

CRediT authorship contribution statement

Jian-Sheng Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Funding acquisition, Formal analysis, Conceptualization; **Wen-Ting Li:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis; **Jun-Yun Wu:** Writing – review & editing, Visualization, Validation, Software, Methodology, Conceptualization; **Weidong Min:** Writing – review & editing, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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